

Learn IN Depth

Be Professional In Embedded System



Project **1** Learn in Depth

PRESSURE DETECTION SYSTEM

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Embedded Systems | 05.08.2024

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Case Study

For this project, the use case is defined as follows: a pressure controller shall inform an airplane cabin crew with an alarm when the pressure will exceed 20 bar.

The alarm duration should be 60 seconds.

A nice to have feature is to keep track or rather save the measured values into the memory.

For this use case a system has to be designed to fulfill this use case.

In order to have a clear definition of the use case, these main assumptions have been taken into consideration:

- The system maintenance is not included.
- The pressure sensor never fails.
- The alarm never fails.
- The system never faces power cut.

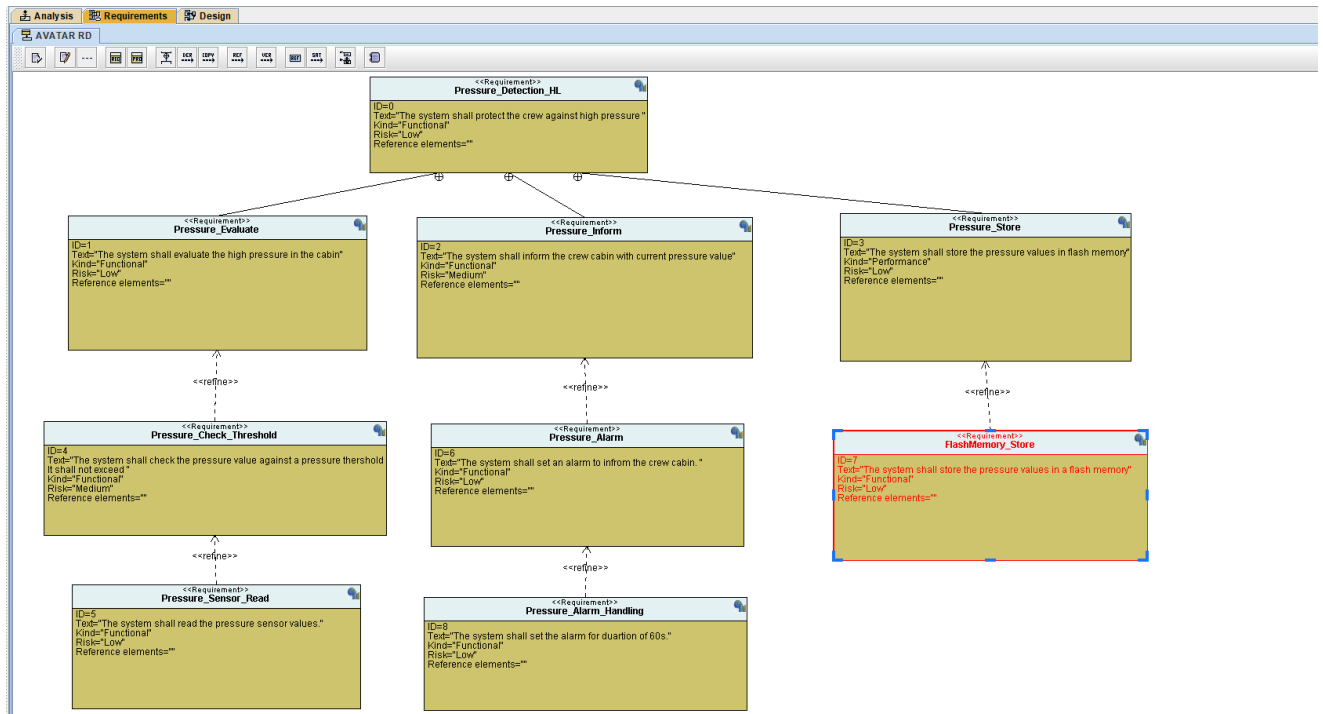
The system setup and shutdown procedures are not modeled.

Methodology

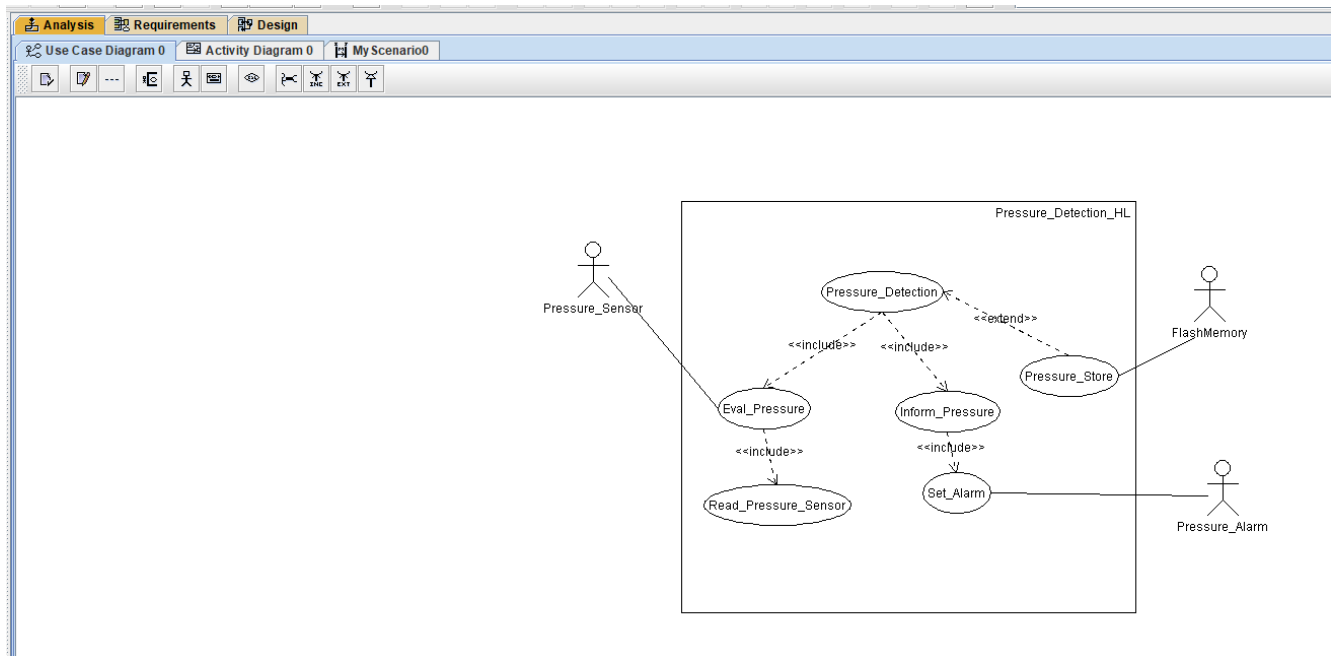
In order to develop a proper system, one of software development life cycle models has to be followed. In this project, the V-Model has been chosen to be used and followed, since it has a lot of advantages that help us have a proper methodology to achieve the project goals.

In this following section, we start from the left side of the V-Model with requirements elicitation and management in way that is clear for all parties (customer & supplier), and then moving forward to the next phases of documentations using Unified Modelling Language UML. This is used as common language, to illustrate the main aspects of the projects till detailed design. Finally, simulations of the entire system have been performed that ensure the proper functionality of the entire software design.

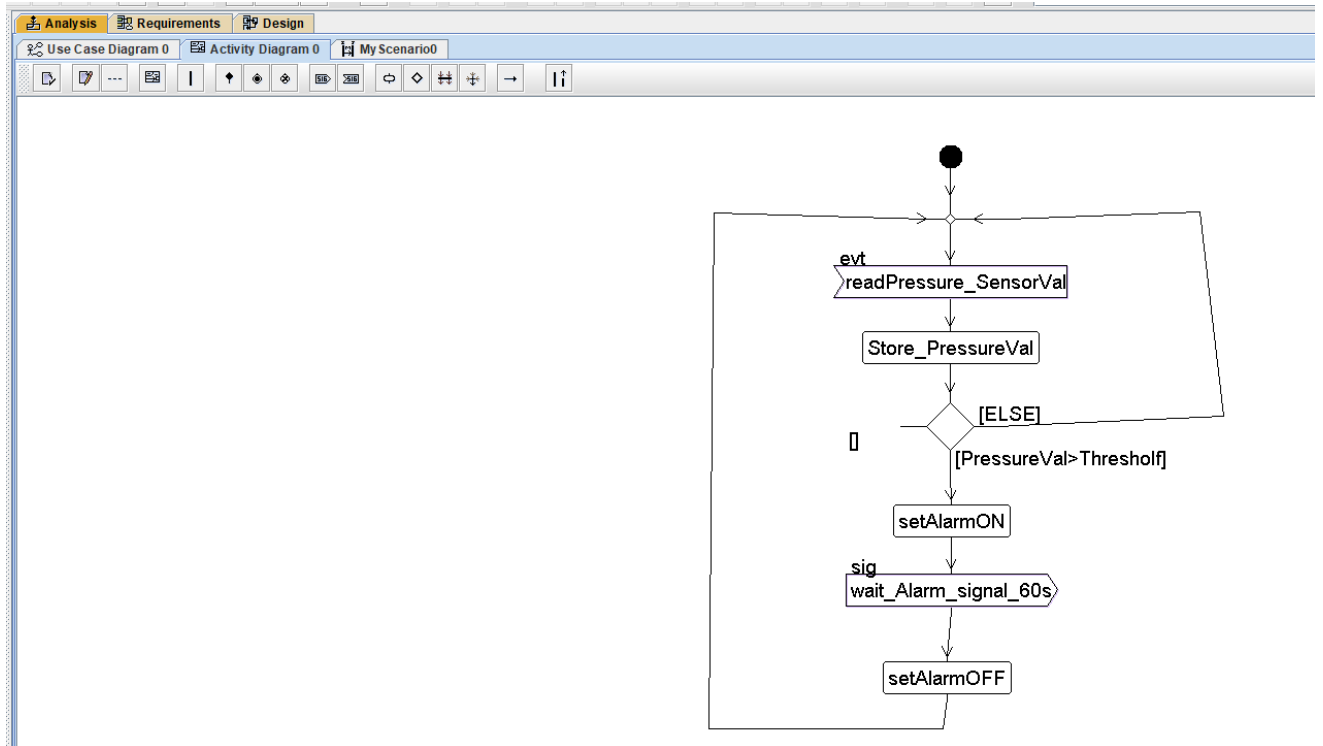
Requirement Diagram



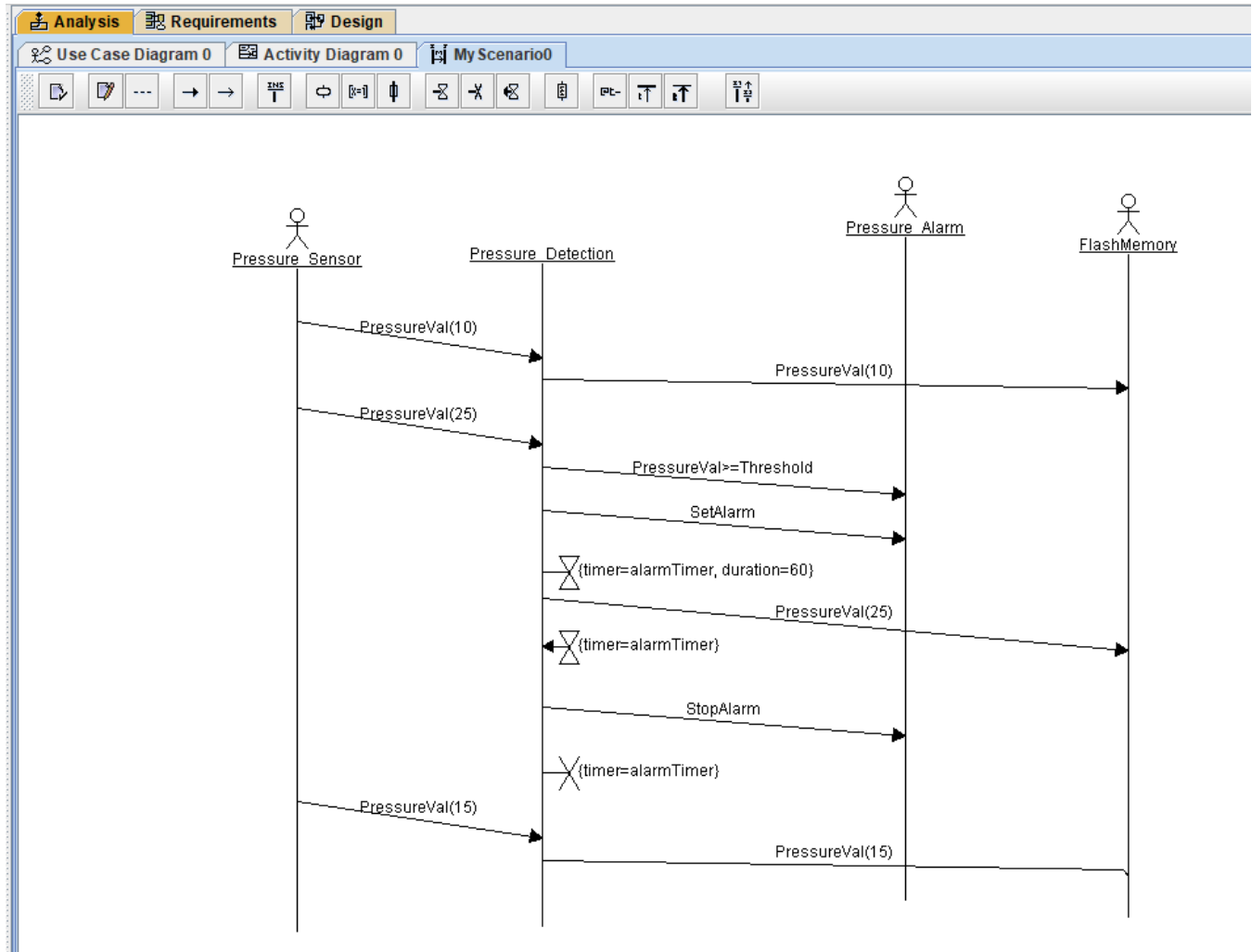
System Analysis / Use Case Diagram



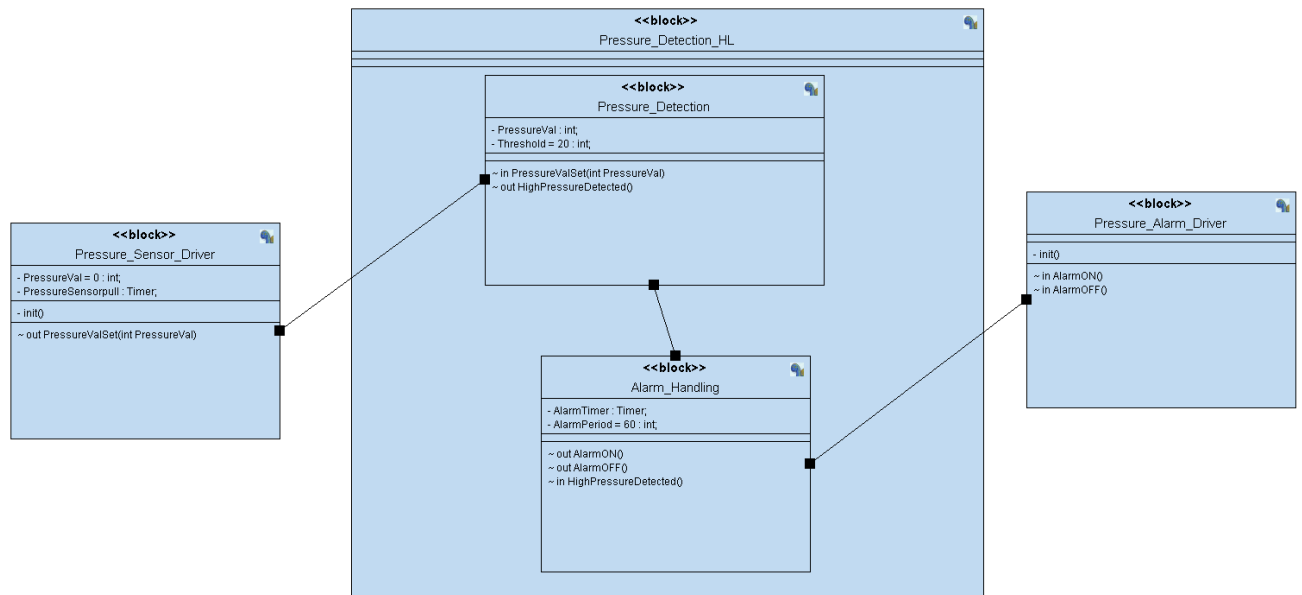
Activity Diagram



Sequence Diagram

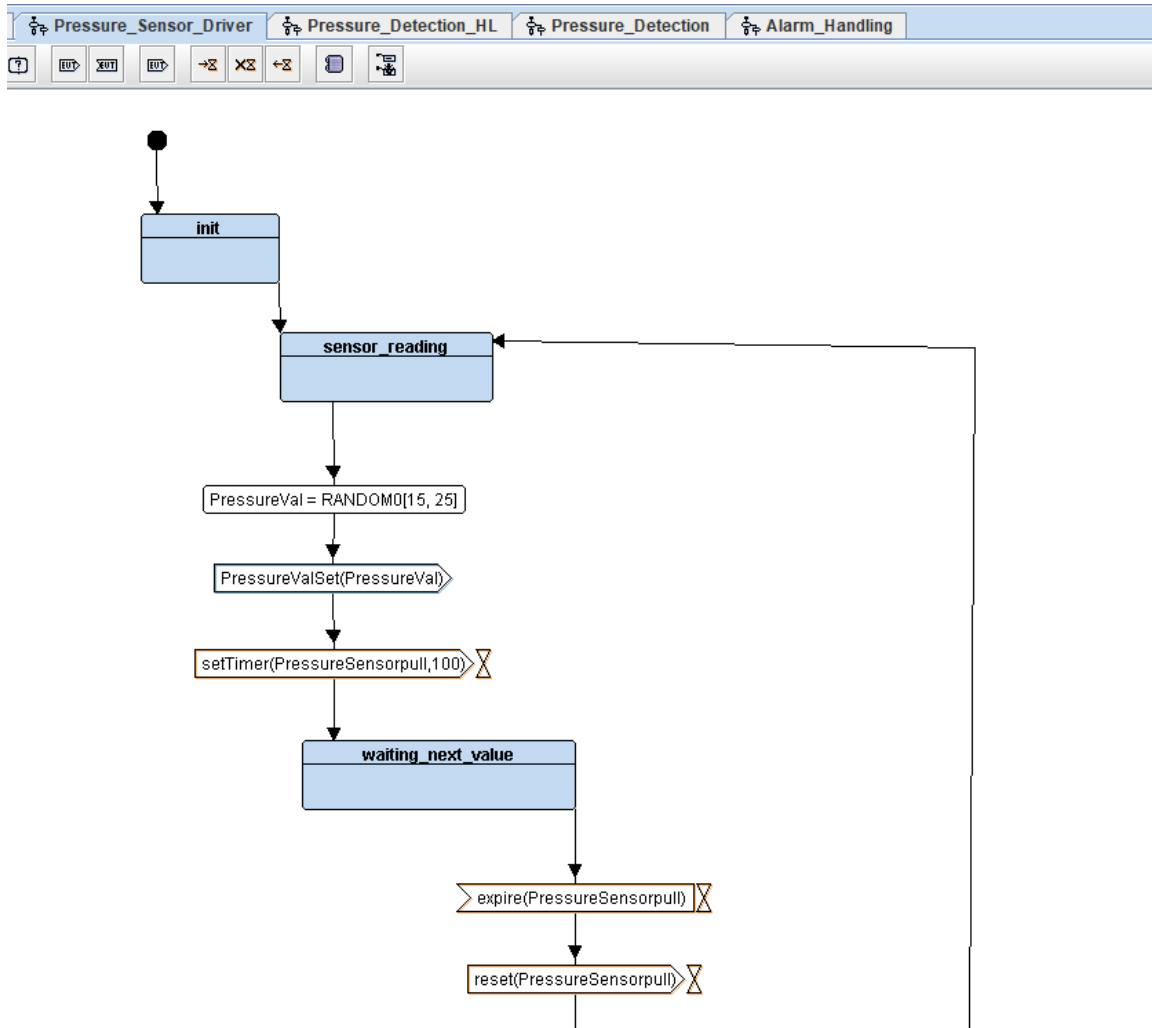


System / Software Design

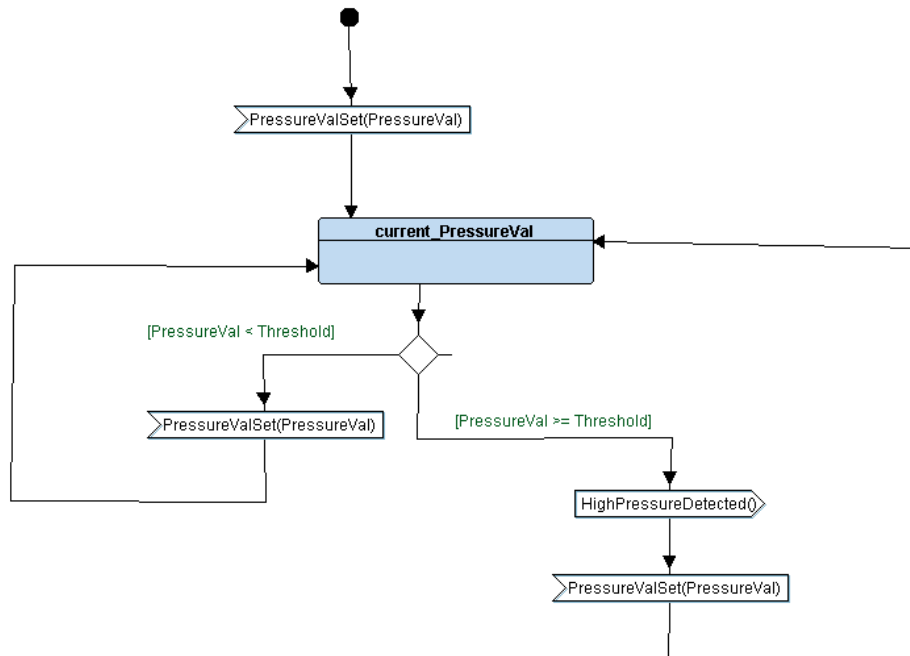
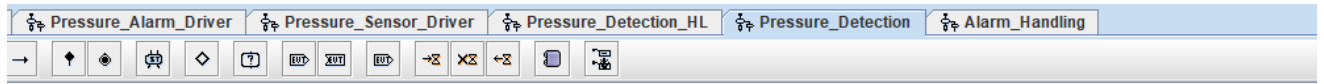


State Diagrams:

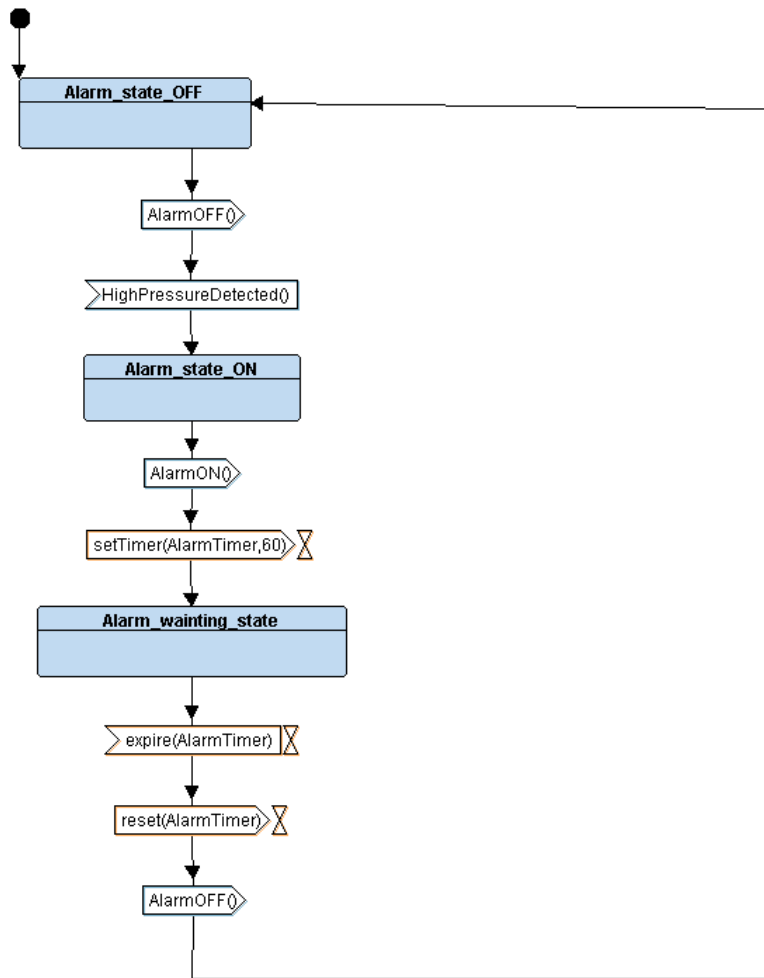
- Pressure Sensor Diver



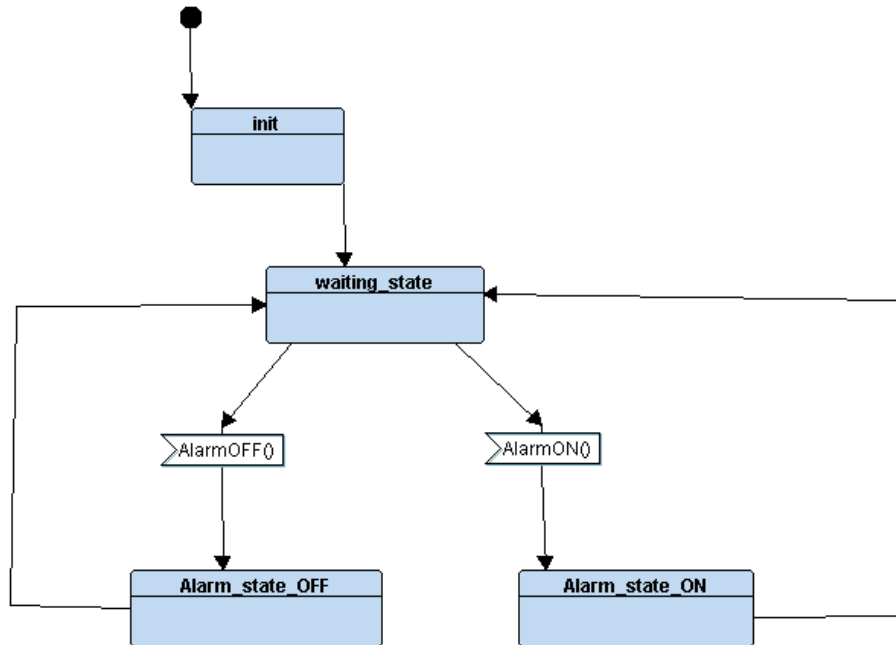
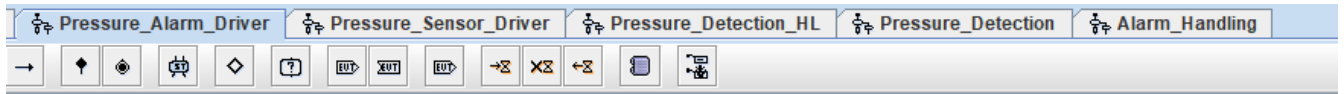
- Pressure Detection Main Functionality



- Alarm handling



- Pressure Alarm Driver



Software Implementation

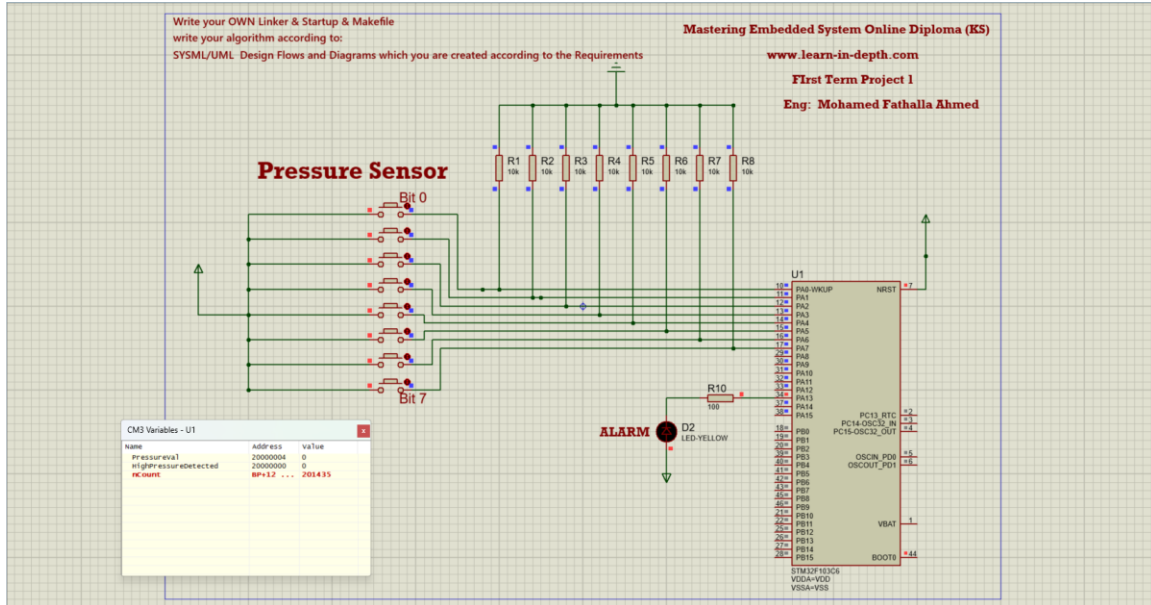
After going through all above diagrams documentation, it was directly started with implementation of each software module as .c & .h files. In the GitHub project, one can find all related files along with startup & linker scripts that are used to bring up the entire software on STM32 Controller.

Symbol Table:

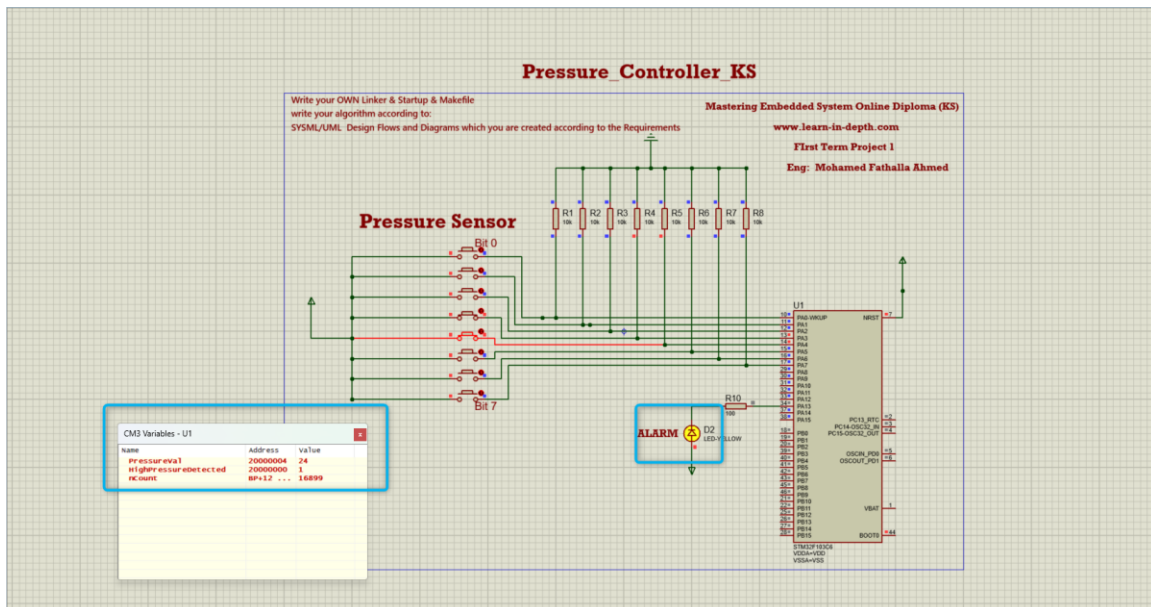
```
$ arm-none-eabi-nm.exe Proj_1_Final_cortexM3.elf
20000008 B _E_bss
20000000 T _E_DATA
080002d8 T _E_TEXT
20000000 B _S_bss
20000000 T _S_DATA
20001008 B _stack_top
0800001c T alarm_handling_func
0800018c T AlarmOFF
0800017c T AlarmON
08000214 W Bus_Fault
080001a8 T current_PressureVal
08000214 T Default_Handler
0800005c T Delay
08000080 T getPressureVal
080000e8 T GPIO_INITIALIZATION
08000214 W H_Fault_Handler
20000000 B HighPressureDetected
0800019c T init_alarm
08000168 T main
08000214 W MM_Fault_Handler
08000214 W NMI_Handler
20000004 B PressureVal
08000220 T Rest_Handler
080001e8 T sensor_reading
08000098 T Set_Alarm_actuator
08000214 W Usage_Fault_Handler
08000000 T vectors
08000200 T waiting_next_value
```

System Verification with Proteus Simulation

Initial State



High Pressure Detected = 1



High Pressure NOT Detected = 0

